

GATE score coefficient from summary-level pQTL statistics

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1 Background

The **GATE** (Genetically Aided Trait Estimation) method constructs a composite genetic score for an exposure by summing the genetically predicted exposure values over all trans-pQTLs, then tests this GATE score for association with a disease outcome. This document derives a formula that computes the log-OR per SD of the GATE score—and its standard error—directly from the per-locus summary statistics $(\hat{\alpha}_k, \hat{\gamma}_k, \text{SE}(\hat{\gamma}_k), \text{sd}_{Z,k})$ that are already available from the MR-Hevo pipeline, without requiring access to individual-level data.

2 Notation

Symbol	Meaning
k	Trans-pQTL locus index; $k = 1, \dots, K$
j	SNP index within locus k
G_{ij}	Genotype (0/1/2 copies of effect allele) for individual i , SNP j
$\hat{\beta}_{m,kj}$	Per-allele instrument weight for SNP j in locus k (from UKBB zarr LD)
$Z_{ik} = \sum_j G_{ij} \hat{\beta}_{m,kj}$	Scalar instrument score for individual i at locus k
$\hat{\alpha}_k$	Instrument-exposure coefficient: $E[\Delta X \mid \Delta Z_k = 1]$
$\hat{\gamma}_k$	Instrument-outcome coefficient: $E[\Delta \log \text{OR}(Y) \mid \Delta Z_k = 1]$
$\text{SE}(\hat{\gamma}_k)$	Standard error of $\hat{\gamma}_k$ (from LD-approximate formula)
$\text{sd}_{Z,k}$	SD of Z_k in the UKBB reference panel (stored as <code>sd.Z</code>)

Symbol	Meaning
N, p	Total sample size and case fraction in the target dataset

3 The GATE score

The GATE score for individual i , summed over all trans-pQTLs, is

$$G_i = \sum_{k=1}^K \hat{\alpha}_k Z_{ik},$$

where the cis-pQTL is excluded. Trans-pQTL loci are in distinct genomic regions and are therefore approximately uncorrelated, so

$$\text{Var}_{\text{ref}}(G) = \sum_k \hat{\alpha}_k^2 \text{sd}_{Z,k}^2$$

with no cross-terms. The standardised GATE score is

$$G_i^* = G_i / \text{SD}_{\text{ref}}(G).$$

4 Score test for $\hat{\beta}_{\text{GATE}}$

4.1 Score and information

Let $Y_i \in \{0, 1\}$ denote disease status and let $\hat{\mu}_i^{(0)}$ denote the fitted probability under the null logistic model (intercept + covariates, no genetic term). Define

$$r_i = Y_i - \hat{\mu}_i^{(0)}, \quad w_i = \hat{\mu}_i^{(0)}(1 - \hat{\mu}_i^{(0)}).$$

The score and partial Fisher information for a scalar instrument X are

$$U_X = \mathbf{x}^\top \mathbf{r}, \quad V_X = \mathbf{x}^\top \mathbf{W} \mathbf{x} - \mathbf{x}^\top \mathbf{W} \mathbf{Z} (\mathbf{Z}^\top \mathbf{W} \mathbf{Z})^{-1} \mathbf{Z}^\top \mathbf{W} \mathbf{x},$$

where $\mathbf{Z}$ is the null-model design matrix (intercept + covariates) and $\mathbf{W} = \text{diag}(w_i)$. The score estimate is $\hat{\beta} = U_X / V_X$ with $\text{SE}(\hat{\beta}) = 1 / \sqrt{V_X}$.

Since the null model includes an intercept and all covariates, the per-locus score $U_k = \mathbf{z}_k^\top \mathbf{r}$ already adjusts for their correlations with Z_k .

4.2 Combining over trans-pQTLs

For the standardised composite $G^* = G / \text{SD}_G = (\sum_k \hat{\alpha}_k Z_k) / \text{SD}_G$:

$$U_{G^*} = \frac{1}{\text{SD}_G} \sum_k \hat{\alpha}_k U_k, \quad V_{G^*} = \frac{1}{\text{SD}_G^2} \sum_k \hat{\alpha}_k^2 V_k,$$

using the independence of instruments across loci ($U_k \perp U_l$, $V_k \perp V_l$ for $k \neq l$).

The score estimate is

$$\hat{\beta}_{\text{GATE}} = \frac{U_{G^*}}{V_{G^*}} = \text{SD}_G \cdot \frac{\sum_k \hat{\alpha}_k U_k}{\sum_k \hat{\alpha}_k^2 V_k}.$$

Substituting $U_k = \hat{\gamma}_k V_k$ (the score estimate for locus k):

$$\hat{\beta}_{\text{GATE}} = \text{SD}_G \cdot \frac{\sum_k \hat{\alpha}_k \hat{\gamma}_k / \text{SE}(\hat{\gamma}_k)^2}{\sum_k \hat{\alpha}_k^2 / \text{SE}(\hat{\gamma}_k)^2}$$

$$\text{SE}(\hat{\beta}_{\text{GATE}}) = \frac{\text{SD}_G}{\sqrt{\sum_k \hat{\alpha}_k^2 / \text{SE}(\hat{\gamma}_k)^2}}$$

5 Simplification of the standard error

The LD-approximate SE formula used by the pipeline (see `genoscores_zarr_loci.py`) is

$$\text{SE}(\hat{\gamma}_k) = \frac{1}{\sqrt{N \text{sd}_{Z,k}^2 p(1-p)}}, \quad \text{so} \quad V_k = \frac{1}{\text{SE}(\hat{\gamma}_k)^2} = N \text{sd}_{Z,k}^2 p(1-p).$$

Substituting into the denominator of (4.2):

$$\sum_k \hat{\alpha}_k^2 V_k = N p(1-p) \sum_k \hat{\alpha}_k^2 \text{sd}_{Z,k}^2 = N p(1-p) \text{SD}_G^2.$$

Therefore

$$\text{SE}(\hat{\beta}_{\text{GATE}}) = \frac{\text{SD}_G}{\sqrt{N p(1-p) \text{SD}_G^2}} = \frac{1}{\sqrt{N p(1-p)}}.$$

The SE is determined entirely by the sample size and case fraction, and is equal to the SE that would be obtained from a single 1-df binary logistic regression.

6 Equivalence to the IVW z-statistic

Substituting $V_k = N \text{sd}_{Z,k}^2 p(1-p)$ into the numerator and defining the per-SD(Z) quantities $\hat{\alpha}_k^* = \hat{\alpha}_k \text{sd}_{Z,k}$ and $\hat{\gamma}_k^* = \hat{\gamma}_k \text{sd}_{Z,k}$ (the values displayed in the instrument tables), the z-statistic becomes

$$z_{\text{GATE}} = \frac{\hat{\beta}_{\text{GATE}}}{\text{SE}(\hat{\beta}_{\text{GATE}})} = \sqrt{N p(1-p)} \cdot \frac{\sum_k (\hat{\alpha}_k^*)^2 \hat{\theta}_k}{\sqrt{\sum_k (\hat{\alpha}_k^*)^2}},$$

where $\hat{\theta}_k = \hat{\gamma}_k^* / \hat{\alpha}_k^*$ is the per-locus IV-ratio estimate. This is the IVW z-statistic with weights $w_k = (\hat{\alpha}_k^*)^2$ proportional to the reciprocal variance of the ratio estimator—identical to the MR-Hevo IVW estimator restricted to trans-pQTLs.

7 Comparison: summary-level vs direct regression (PDCD1/T1D)

The summary-level formula is validated against the directly computed coefficient from logistic regression of T1D on the individual-level GATE score in the SDRNT1BIO/GS dataset (4,922 cases, 7,452 controls). For the direct computation, the GATE score is accumulated as $G_i = \sum_{k,j} G_{ij} \cdot (\hat{\alpha}_k \hat{\beta}_{m,kj})$ across all trans-pQTL SNPs in the individual-level genotype data on `diabepi`.

The two methods differ in the SD used for standardisation: the summary-level formula uses $\text{SD}_{\text{ref}}(G)$ from the UKBB reference panel, while the direct regression standardises by the empirical SD in the target GS dataset. The z-statistics are invariant to this scaling difference; the ratio $\hat{\beta}_{\text{summary}} / \hat{\beta}_{\text{direct}}$ estimates $\text{SD}_{\text{ref}}(G) / \text{SD}_{\text{target}}(G)$ and should be close to 1 for a well-calibrated reference panel.

Table 2: GATE coefficient for T1D ($\hat{\beta}_{\text{GATE}} = \text{log-OR per SD of GATE score}$) from summary-level formula vs direct individual-level regression. Trans-pQTLs: number of pQTL loci contributing to the GATE score (cis-pQTL PDCD1 excluded). SE (summary-level) = $1/\sqrt{Np(1-p)} \approx 0.0090$; SE (direct) uses the empirical SD of the GATE score in the GS target dataset.

Method	$\hat{\beta}$	SE	z	p	Trans pQTLs
Summary-level	0.2244	0.01837	12.218	2.5e-34	38
Direct (individual-level)	0.1980	0.02044	9.687	3.42e-22	41

The z-statistics from both methods should agree closely. Any discrepancy reflects LD mismatch between the UKBB reference panel and the GS target dataset, analogous to the attenuation seen in the LD-approximate $\hat{\gamma}$ estimates (Pearson $r \approx 0.83$ in the Analysis~2 vs Analysis~3 comparison in the main report).